MAT11-1

Foundations

Logic, Numbers, Functions, and the Calculus

The analytical and algebraic toolkit that every later course will quietly assume you own.

Albert School — 38 hours

Preface

You are not arriving here empty-handed. From high school you already know how to solve a quadratic, sketch a parabola, differentiate x^2, and read off the sine of an angle. This course does not replace that knowledge; it **re-founds** it. The same objects — numbers, functions, derivatives — return, but this time we ask the questions a mathematician asks: **what exactly is being claimed, and why is it true?**

That shift is the whole point of a "Foundations" course. In high school a theorem was a formula to apply. Here a theorem is a precise sentence with hypotheses and a conclusion, and your job is to **read** it, **negate** it, **prove** it, and know exactly when it does and does not apply. The tools you build — propositional logic, set notation, the real-number system, the careful definitions of limit, continuity, derivative, and integral — are the vocabulary in which every subsequent mathematics, statistics, and machine-learning course will be written. If a later lecturer writes "let f be continuous on a compact interval, then…", they assume that sentence already means something precise to you. Making it mean something precise is what we do now.

The course runs in the natural order of dependence. We start with **logic and notation**, because we cannot state anything carefully without them. We build the **real numbers**, because they are the stage on which everything happens. We study **functions** and **polynomials** as the first rich objects, **sequences** as the first place formal convergence appears, and then the central pillars of calculus: **limits and continuity**, **derivatives**, and **integration**. We close with **complex numbers**, which complete the number system and reveal a startling unity between exponentials and trigonometry.

A word on how to read this book. Every new idea is introduced with a concrete example **before** it is stated abstractly — meet the animal before you classify it. Boxes labelled **Common pitfall** flag the specific mistakes students make every year; read them as if they were written about you, because they were. Work the exercises. Mathematics is not a spectator sport, and reading a proof is to constructing one as watching someone swim is to swimming.

Contents

1 The Language of Logic

1.1 A theorem you cannot trust until you can read it

Consider the claim: **“If a whole number n is divisible by 6, then it is divisible by 3.”** You believe it. Now consider its **converse**: **“If n is divisible by 3, then it is divisible by 6.”** That is false ($n = 9$). Same two properties, the arrow reversed, and truth flips to falsehood. Most reasoning errors in mathematics — and in life — are exactly this: confusing a statement with one of its rearrangements. This chapter builds the machinery that makes such confusions impossible.

1.2 Propositions and connectives

Definition 1 (Proposition) A **proposition** is a statement that is either **true** or **false**, but not both. “$5 > 3$” is a proposition (true). “$x > 3$” is **not** a proposition until we know x — it is a **predicate**, a statement with a free variable, which becomes a proposition once the variable is fixed or quantified.

From propositions P and Q we build compound propositions using **connectives**. Each is defined by a **truth table**, which lists the truth value of the compound for every combination of inputs.

Definition 2 (The basic connectives)
- **Negation** $\neg P$ (“not P”) is true exactly when P is false.
- **Conjunction** $P \wedge Q$ (“P and Q”) is true exactly when both are true.
- **Disjunction** $P \vee Q$ (“P or Q”) is true when at least one is true. This is the **inclusive** or: it is **true** when both hold.
- **Implication** $P \Rightarrow Q$ (“P implies Q”, “if P then Q”) is false **only** in the single case P true and Q false.
- **Equivalence** $P \Leftrightarrow Q$ (“P if and only if Q”) is true exactly when P and Q have the same truth value.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Table 1: The truth tables of the five connectives, all on one grid. Read each row as one possible world; the column tells you whether the compound is true in that world.

The implication row deserves a stare. Why is “$P \Rightarrow Q$” **true** whenever P is false? Think of a promise: **“If it rains, I will bring an umbrella.”** You have broken that promise only on a day when it rains and you have no umbrella. On a dry day you have broken nothing — whatever you carry. An implication with a false hypothesis is **vacuously true**; it makes no claim.

Common pitfall “$P \Rightarrow Q$” does **not** mean P causes Q, nor that P and Q are both true. “If $2 + 2 = 5$, then I am the Pope” is a **true** implication, because its hypothesis is false. Implication is about the **pattern of truth values**, not about meaning or causation.

1.3 Contrapositive, converse, and the mistakes they breed

Given "$P \Rightarrow Q$", three relatives lurk:

Definition 3 (Converse, inverse, contrapositive) For the implication $P \Rightarrow Q$:

- its **converse** is $Q \Rightarrow P$;
- its **inverse** is $\neg P \Rightarrow \neg Q$;
- its **contrapositive** is $\neg Q \Rightarrow \neg P$.

The single most useful fact in this chapter:

Theorem 4 (An implication equals its contrapositive) $P \Rightarrow Q$ and $\neg Q \Rightarrow \neg P$ are **logically equivalent** — they have identical truth tables. By contrast, the converse and inverse are **not** equivalent to the original (though they are equivalent to each other).

Proof. Compare columns. $P \Rightarrow Q$ is false only when P is T and Q is F. $\neg Q \Rightarrow \neg P$ is false only when $\neg Q$ is T and $\neg P$ is F, i.e. Q is F and P is T — the very same row. Two propositions false on exactly the same rows are true on exactly the same rows, hence equivalent. □

Worked example — Proving by contrapositive. Claim: **"If n^2 is even, then n is even."** Proving this directly is awkward. The contrapositive — **"if n is odd, then n^2 is odd"** — is easy: if $n = 2k+1$ then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. Since the contrapositive is true and equivalent to the original, the original is proved.

Common pitfall **Affirming the converse** is the classic blunder. From "if it rained, the ground is wet" and "the ground is wet", you may **not** conclude "it rained" — someone may have used a hose. The truth of $P \Rightarrow Q$ tells you nothing about $Q \Rightarrow P$.

1.4 Negating compound statements

To disprove a claim, or to run a proof by contradiction, you must negate it correctly. The rules are mechanical once memorised.

Proposition 5 (De Morgan's laws and friends)

$$\begin{aligned} \neg(P \wedge Q) &\Leftrightarrow (\neg P) \vee (\neg Q), \\ \neg(P \vee Q) &\Leftrightarrow (\neg P) \wedge (\neg Q), \\ \neg(P \Rightarrow Q) &\Leftrightarrow P \wedge (\neg Q). \end{aligned}$$

The last one is the most often botched. The negation of "if P then Q" is **not** another implication — it is the assertion that P holds **and yet** Q fails. To deny the promise "if it rains I bring an umbrella" is to describe one rainy, umbrella-less day.

Worked example — Negating a quantified statement. We will meet quantifiers properly in the next chapter, but the pattern is worth seeing now. The negation of "**every** student passed" is "**some** student did not pass" — not "every student failed". Negation swaps "for all" with "there exists" and negates the inside.

1.5 Analysis–synthesis: reasoning toward existence

Many problems ask: **does a solution exist, and if so, which?** A powerful two-phase method handles these.

Definition 6 (Analysis–synthesis)

- **Analysis: assume** a solution exists and derive necessary conditions it must satisfy. This narrows the candidates down — often to a single one — but proves nothing yet, because the assumption might be empty.
- **Synthesis:** take each surviving candidate and **verify** it actually solves the problem. Only verified candidates are genuine solutions.

The two phases answer different questions: analysis answers "**what could** a solution be?", synthesis answers "**which of those really are**?". Skipping synthesis is a real error, not a formality — analysis can produce phantom candidates that fail on substitution.

Worked example — Splitting a function into even and odd parts. Claim: every function $f : \mathbb{R} \to \mathbb{R}$ can be written uniquely as $f = e + o$ with e even ($e(-x) = e(x)$) and o odd ($o(-x) = -o(x)$).

Analysis. Suppose such e, o exist. Then $f(x) = e(x) + o(x)$ and $f(-x) = e(x) - o(x)$. Adding and subtracting forces

$$e(x) = \frac{f(x) + f(-x)}{2}, \quad o(x) = \frac{f(x) - f(-x)}{2}.$$

So **if** a decomposition exists, it must be this one — uniqueness is settled.

Synthesis. Define e, o by those formulas and check: e is even, o is odd, and $e + o = f$. They work. Existence is settled. The two phases together prove existence **and** uniqueness.

Exercises

1. Build the truth table of $(P \Rightarrow Q) \wedge (Q \Rightarrow P)$ and confirm it equals the table of $P \Leftrightarrow Q$.
2. Write the contrapositive, converse, and inverse of "if a function is differentiable, then it is continuous." Which are true?
3. Negate, simplifying so that $\neg$ touches only atomic statements: $\neg(P \vee (\neg Q))$ and $\neg(P \Rightarrow (Q \wedge R))$.
4. Using analysis–synthesis, find all real x with $\sqrt{x+2} = x$. (Note how synthesis discards a candidate the analysis produces.)

Chapter takeaways. A proposition is true or false; connectives are defined by truth tables. An implication is false only when its hypothesis is true and conclusion false, and equals its contrapositive — never its converse. Negation follows De Morgan, and $\neg(P \Rightarrow Q)$ is $P \wedge \neg Q$. Analysis–synthesis proves existence/uniqueness by deriving necessary conditions, then verifying the survivors.

2 Sets, Notation, and Reading Mathematics

2.1 The need for a precise notation

A theorem is only as clear as the symbols it is written in. "The set of points closer to the origin than to $(1, 0)$" is a sentence; $\{(x, y) : x^2 + y^2 < (x - 1)^2 + y^2\}$ is a **computation waiting to happen**. This chapter is the dictionary that lets you translate between the two — in both directions, because **reading** formal mathematics and **writing** it are the two halves of literacy.

2.2 Sets and membership

Definition 7 (Set notation) A **set** is a collection of objects, its **elements**.

- $x \in A$: "x is an element of A"; $x \notin A$ denies it.
- **Builder notation** $\{x \in A : P(x)\}$: the elements of A satisfying the predicate P. Read the colon as "such that".
- $A \subset B$ ("A is a subset of B"): every element of A is in B, i.e. $x \in A \Rightarrow x \in B$.
- **Union** $A \cup B = \{x : x \in A \vee x \in B\}$.
- **Intersection** $A \cap B = \{x : x \in A \wedge x \in B\}$.
- The **empty set** $\varnothing$ has no elements.

Notice the connectives from Chapter 1 reappearing: union is built on "or", intersection on "and", subset on implication. Set theory is logic wearing a different coat.

The standard number systems, nested by inclusion:

$$\mathbb{N} \subset \mathbb{Z} \subset \mathbb{Q} \subset \mathbb{R} \subset \mathbb{C},$$

the naturals, integers, rationals, reals, and complex numbers — each containing the last.

> **Worked example — Reading a set out loud.** $A = \{n \in \mathbb{N} : n \text{ is even and } n \leq 10\}$ reads "the set of natural numbers n such that n is even and at most 10", i.e. $\{0, 2, 4, 6, 8, 10\}$. Translating **to** notation: "the rationals strictly between 0 and 1" becomes $\{q \in \mathbb{Q} : 0 < q < 1\}$.

2.3 Quantifiers

Definition 8 (Universal and existential quantifiers)

- $\forall x \in A,\ P(x)$: "**for all** x in A, $P(x)$ holds."
- $\exists x \in A,\ P(x)$: "**there exists** an x in A with $P(x)$."
- $\exists! x \in A,\ P(x)$: "there exists a **unique** such x."

Common pitfall **Order of quantifiers changes meaning.** "$\forall x, \exists y : y > x$" (for every number there is a bigger one — true over $\mathbb{R}$) is utterly different from "$\exists y, \forall x : y > x$" (one number bigger than all — false). When you swap $\forall$ and $\exists$, you have written a new statement, usually with a new truth value.

To negate quantifiers, push the negation inward, flipping each quantifier:

$$\neg(\forall x,\ P(x)) \Leftrightarrow \exists x,\ \neg P(x), \quad \neg(\exists x,\ P(x)) \Leftrightarrow \forall x,\ \neg P(x).$$

2.4 Summation notation

Definition 9 (The summation symbol)

$$\sum_{i=1}^{n} a_i = a_1 + a_2 + \cdots + a_n.$$

The **index** i runs over consecutive integers from the lower to the upper bound. It is a **dummy**: $\sum_{i=1}^{n} a_i$ and $\sum_{j=1}^{n} a_j$ are the same number. An empty sum (upper bound below lower) is 0 by convention.

Worked example — Two sums worth knowing.

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2}, \qquad \sum_{i=0}^{n} r^i = \frac{1-r^{n+1}}{1-r} \quad (r \neq 1).$$

The first is the triangular-number formula; the second is the **finite geometric sum**, which will drive the whole theory of geometric sequences in Chapter 6.

2.5 Reading a theorem statement

Putting it together, here is how a working mathematician parses a theorem. Take: **"For every $\varepsilon > 0$ there exists $\delta > 0$ such that for all x, $|x - a| < \delta \Rightarrow |f(x) - f(a)| < \varepsilon$."**

Decode it phrase by phrase: it opens with $\forall \varepsilon$, so the claim must hold **no matter how small** a tolerance ε someone picks; then $\exists \delta$, so **we** get to choose a δ in response, possibly depending on ε; then $\forall x$ an implication, the actual guarantee. We will meet this exact sentence again as the definition of continuity — and you will already know how to read it.

Exercises

5. Translate to notation: "the set of real x whose square is at most 2" and "every integer is either even or odd".
6. Compute $\sum_{k=2}^{5}(2k-1)$ and $\sum_{i=1}^{4} i^2$.
7. Negate, pushing $\neg$ all the way in: $\forall \varepsilon > 0, \exists N, \forall n \geq N : |a_n - L| < \varepsilon$.
8. Are $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$ for all sets? Argue using the logic of "and" over "or".

Chapter takeaways. Sets are described by membership and builder notation; union/intersection/subset mirror or/and/implication. Quantifiers $\forall, \exists$ do not commute, and negating one flips it. Summation $\sum$ uses a dummy index. Reading a theorem means decoding its quantifiers in order — who chooses what, and in what dependence.

3 The Real Numbers

3.1 Why the rationals are not enough

The Greeks discovered, to their dismay, that no fraction squares to 2: if $\frac{p^2}{q^2} = 2$ in lowest terms, then $p^2 = 2q^2$ is even, so p is even, so p^2 is divisible by 4, forcing q^2 — and hence q — even too, contradicting "lowest terms". So $\sqrt{2} \notin \mathbb{Q}$. The rational number line is full of **holes**. The real numbers $\mathbb{R}$ are the system that fills them, and almost everything in calculus depends on its having no gaps.

3.2 The ordered field axioms

Definition 10 (Ordered field) $\mathbb{R}$ is an **ordered field**: it carries addition and multiplication satisfying the familiar laws — associativity, commutativity, distributivity, an additive identity 0 and inverses $-x$, a multiplicative identity 1 and inverses x^{-1} for $x \neq 0$ — together with a total order $<$ compatible with the operations:

- if $a < b$ then $a + c < b + c$ for all c;
- if $a < b$ and $c > 0$ then $ac < bc$.

Every algebraic move you make — clearing denominators, multiplying an inequality through — is licensed by one of these axioms. The second order axiom carries the famous warning:

Common pitfall Multiplying an inequality by a **negative** number reverses it: from $a < b$ and $c < 0$ we get $ac > bc$. Forgetting this flip is the most common error in solving inequalities. $-2x < 6$ gives $x > -3$, not $x < -3$.

3.3 Completeness: the line with no gaps

What truly separates $\mathbb{R}$ from $\mathbb{Q}$ is one further property.

Definition 11 (Completeness (informal)) $\mathbb{R}$ is **complete**: it has no gaps. Precisely, every non-empty set of reals that is bounded above has a **least upper bound** (supremum) in $\mathbb{R}$. The set $\{q \in \mathbb{Q} : q^2 < 2\}$ is bounded above but has no least upper bound **in** $\mathbb{Q}$ — its would-be bound is $\sqrt{2}$, which is missing. In $\mathbb{R}$ the bound exists.

Remark We **state** completeness and use it; the fully formal construction of $\mathbb{R}$ is deferred to a later analysis course (AD, Semester 3). For now, "no gaps" is the working intuition: any point you can squeeze between approaching values is actually there. This is what guarantees that $\sqrt{2}$, π, limits, and the Intermediate Value Theorem's roots all genuinely exist.

3.4 Intervals, absolute value, distance

Definition 12 (Intervals) For $a \leq b$: the **closed** interval $[a, b] = \{x : a \leq x \leq b\}$ includes its endpoints; the **open** interval $(a, b) = \{x : a < x < b\}$ excludes them; half-open $[a, b)$, $(a, b]$ mix. Unbounded ones use ∞ as a **symbol**, never a number: $[a, \infty) = \{x : x \geq a\}$.

Definition 13 (Absolute value and distance)

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

Geometrically $|x|$ is the distance from x to 0, and $|x - y|$ is the **distance between x and y** on the line. Hence $|x - a| < r$ describes exactly the open interval $(a - r, a + r)$: the points within r of a.

That reading — $|x - a| < r$ as "x is within r of a" — is the single most useful sentence in all of calculus. Every limit and continuity statement is secretly about such neighbourhoods.

3.5 The triangle inequality

Theorem 14 (Triangle inequality) For all real x, y,

$$|x+y| \le |x| + |y|,$$

with equality exactly when x and y have the same sign (or one is zero).

Proof. For any t, $-|t| \le t \le |t|$. Adding the versions for x and y gives $-(|x|+|y|) \le x+y \le |x|+|y|$, which is precisely $|x+y| \le |x|+|y|$. $\square$

The name comes from geometry: the direct path is no longer than a detour. A constantly used variant bounds a difference:

$$|x-z| = |(x-y)+(y-z)| \le |x-y| + |y-z|,$$

"the distance from x to z is at most the distance via y". This is the engine behind almost every ε-estimate later in the book.

Worked example — Bounding with the triangle inequality. Suppose $|x-3| < 0.1$. How large can $\left|x^2-9\right|$ be? Write $\left|x^2-9\right| = |x-3| \cdot |x+3|$. The first factor is < 0.1. For the second, $|x+3| \le |x-3| + |6| < 0.1 + 6 = 6.1$. Hence $\left|x^2-9\right| < 0.1 \times 6.1 = 0.61$. We **bounded** an expression we could not compute exactly — the recurring move of analysis.

Exercises

9. Solve $|2x-1| \le 5$ and write the solution as an interval.
10. Show $||x|-|y|| \le |x-y|$ (the **reverse** triangle inequality). Hint: apply the triangle inequality to $x = (x-y)+y$.
11. Explain why $\{q \in \mathbb{Q} : q^2 < 2\}$ has no least upper bound in $\mathbb{Q}$ but does in $\mathbb{R}$.
12. If $|x-2| < 0.01$, bound $\left|x^3-8\right|$ from above.

Chapter takeaways. $\mathbb{R}$ is an ordered field (multiplying an inequality by a negative flips it) that is moreover **complete** — no gaps, every bounded set has a supremum — which $\mathbb{Q}$ is not. Absolute value measures distance, so $|x-a| < r$ is a neighbourhood of a, and the triangle inequality $|x+y| \le |x|+|y|$ is the basic tool for bounding expressions.

4 Functions

4.1 A machine, then a definition

Think of a function as a machine: drop in an input, exactly one output comes out. The squaring machine sends $3 \mapsto 9$ and $-3 \mapsto 9$; it never sends 3 to two different numbers. That "exactly one output" is the whole idea.

Definition 15 (Function, domain, codomain) A **function** $f : A \to B$ assigns to each element x of the **domain** A exactly one element $f(x)$ of the **codomain** B. The **range** (or image of the whole domain) is $\{f(x) : x \in A\} \subset B$ — the outputs actually attained, which may be a proper subset of B.

Common pitfall **Range and codomain are different.** $f : \mathbb{R} \to \mathbb{R}$, $f(x) = x^2$ has codomain $\mathbb{R}$ but range $[0, \infty)$ — negative numbers are never outputs. Many "is this surjective?" questions are really asking whether range fills the codomain.

4.2 Image and preimage of sets

Definition 16 (Image and preimage) For $f : A \to B$, a subset $S \subset A$, and a subset $T \subset B$:
- the **image** $f(S) = \{f(x) : x \in S\}$ — where S lands;
- the **preimage** $f^{-1}(T) = \{x \in A : f(x) \in T\}$ — everything that lands in T.

The preimage notation f^{-1} here needs **no** inverse function to exist; it is defined for any f.

Worked example — Image and preimage of squaring. For $f(x) = x^2$: the image $f([1, 2]) = [1, 4]$. The preimage $f^{-1}([1, 4]) = [-2, -1] \cup [1, 2]$ — note it is **bigger** than $[1, 2]$, because negatives also square into $[1, 4]$. Preimage and image are not inverse operations in general.

4.3 Composition

Definition 17 (Composition) Given $f : A \to B$ and $g : B \to C$, the **composite** $g \circ f : A \to C$ is $(g \circ f)(x) = g(f(x))$ — do f first, then g. The order matters: $g \circ f$ and $f \circ g$ usually differ.

Worked example — Order matters. With $f(x) = x + 1$ and $g(x) = x^2$: $(g \circ f)(x) = (x + 1)^2$ but $(f \circ g)(x) = x^2 + 1$. Squaring-after-shifting is not shifting-after-squaring.

4.4 Injective, surjective, bijective

These three words classify functions by **how many inputs hit each output**.

Definition 18 (Injective, surjective, bijective) $f : A \to B$ is
- **injective** (one-to-one) if different inputs give different outputs: $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ (no output is hit twice);
- **surjective** (onto) if every $b \in B$ is some $f(x)$ (the range is all of B);
- **bijective** if both — a perfect pairing of A with B.

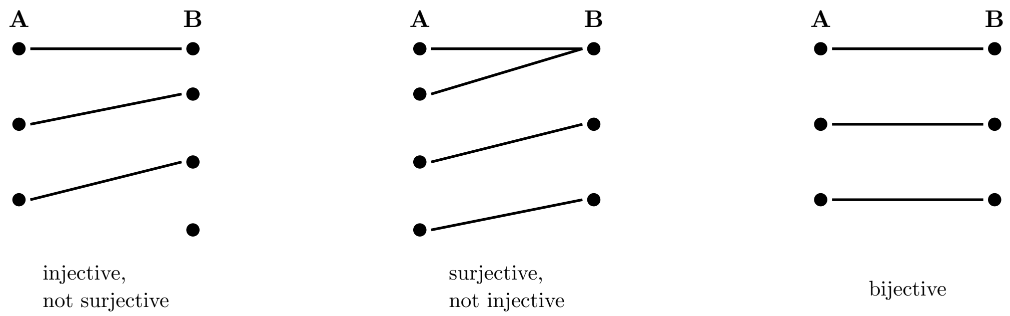

Figure 1: The three notions as arrow diagrams. Injective: no two arrows share a head. Surjective: every right-hand dot is hit. Bijective: a perfect matching.

Remark Surjectivity depends on the **codomain** you declare. $f(x) = x^2$ is not surjective as $\mathbb{R} \to \mathbb{R}$, but **is** surjective as $\mathbb{R} \to [0, \infty)$. Shrinking the codomain to the range always makes a function surjective; restricting the domain can make it injective.

4.5 Inverse functions

Definition 19 (Inverse function) If $f : A \to B$ is **bijective**, its **inverse** $f^{-1} : B \to A$ is the function that undoes it: $f^{-1}(y) = x$ exactly when $f(x) = y$. It satisfies $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$. A function has an inverse **if and only if** it is bijective.

Graphically, $y = f^{-1}(x)$ is the reflection of $y = f(x)$ across the line $y = x$ — input and output swap roles.

Worked example — Restricting to invert. $f(x) = x^2$ is not invertible on $\mathbb{R}$ (not injective). Restricted to $[0, \infty)$ it becomes bijective onto $[0, \infty)$, with inverse $f^{-1}(y) = \sqrt{y}$. We will do exactly this trick to invert the trigonometric functions below.

4.6 Standard families: exponentials and logarithms

Definition 20 (Exponential and logarithm) For base $a > 0$, $a \neq 1$, the **exponential** $x \mapsto a^x$ is defined for all real x, is positive, and satisfies $a^{x+y} = a^x a^y$. It is a bijection $\mathbb{R} \to (0, \infty)$, and its inverse is the **logarithm** $\log_a : (0, \infty) \to \mathbb{R}$, so

$$a^{\log_a y} = y, \qquad \log_{a(a^x)} = x.$$

The **natural** base is $e \approx 2.718$, with $\exp(x) = e^x$ and $\ln = \log_e$. Logarithms turn products into sums: $\ln(xy) = \ln x + \ln y$.

Common pitfall $\ln(x + y)$ is **not** $\ln x + \ln y$. The logarithm law is about **products**, not sums. This false "linearity" is among the most common algebra errors; there is no useful simplification of $\ln(x + y)$.

4.7 Standard families: trigonometry and the unit circle

Trigonometry is best founded not on triangles but on the **unit circle**.

Definition 21 (Sine and cosine via the unit circle) Let θ be an angle measured in **radians** (so a full turn is 2π). The point on the unit circle $x^2 + y^2 = 1$ reached by rotating $(1, 0)$ through θ counterclockwise has coordinates $(\cos\theta, \sin\theta)$. Thus $\cos\theta$ and $\sin\theta$ are **defined** as those coordinates, and $\tan\theta = \frac{\sin\theta}{\cos\theta}$ where $\cos\theta \neq 0$.

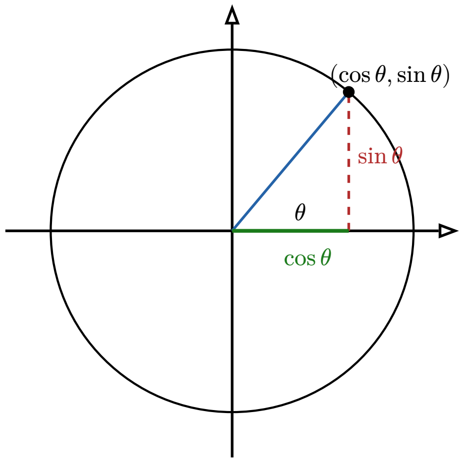

Figure 2: The unit-circle definition. As the point travels round the circle, its horizontal coordinate traces $\cos\theta$ and its vertical coordinate traces $\sin\theta$ — which is why both are periodic with period 2π and bounded in $[-1, 1]$.

From the defining equation $x^2 + y^2 = 1$ we read off, for free, the most important identity in trigonometry, plus the symmetries:

$$\cos^2\theta + \sin^2\theta = 1, \quad \cos(-\theta) = \cos\theta, \quad \sin(-\theta) = -\sin\theta.$$

The **addition formulas** (worth memorising) are

$$\sin(\alpha + \beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta,$$
$$\cos(\alpha + \beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta.$$

Common pitfall Always work in **radians** in calculus. The clean derivative facts $\sin' = \cos$ and $\lim_{x\to 0} \frac{\sin x}{x} = 1$ hold only in radians; in degrees they acquire ugly constant factors. Switch your calculator and your brain to radians and leave them there.

4.8 Inverse trigonometric functions

Sine is not injective on $\mathbb{R}$ (it repeats), so to invert it we **restrict**.

Definition 22 (Inverse trig functions) Restricting sin to $[-\frac{\pi}{2}, \frac{\pi}{2}]$ makes it a bijection onto $[-1, 1]$; its inverse is $\arcsin : [-1, 1] \to [-\frac{\pi}{2}, \frac{\pi}{2}]$. Restricting cos to $[0, \pi]$ gives $\arccos : [-1, 1] \to [0, \pi]$. Restricting tan to $(-\frac{\pi}{2}, \frac{\pi}{2})$ gives $\arctan : \mathbb{R} \to (-\frac{\pi}{2}, \frac{\pi}{2})$.

Common pitfall $\arcsin(\sin x)$ equals x **only** for $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$. For instance $\arcsin(\sin(3\pi)) = \arcsin(0) = 0 \neq 3\pi$. The inverse undoes the function only on the restricted domain where it was made injective.

Exercises

13. For $f(x) = 3x - 4$, decide injectivity and surjectivity as $f : \mathbb{R} \to \mathbb{R}$, and find f^{-1}.
14. With $f(x) = x^2$, compute $f([-1, 2])$ and $f^{-1}([1, 9])$.
15. Let $f(x) = \frac{x}{1+|x|}$ on $\mathbb{R}$. Show its range is $(-1, 1)$ and that it is a bijection onto that range.
16. Simplify $\cos(\arcsin x)$ for $x \in [-1, 1]$. (Draw a right triangle or use $\sin^2 + \cos^2 = 1$.)

17. Using the addition formula, derive $\sin(2\theta) = 2\sin\theta\cos\theta$.

Chapter takeaways. A function sends each input to one output; range (attained outputs) may be smaller than codomain. Image and preimage act on sets and are not mutual inverses. Injective/surjective/bijective count preimages; only bijections have inverses, obtained graphically by reflecting across $y = x$. Exponentials invert to logarithms (ln of a **product** splits), and trig functions, defined on the unit circle in radians, invert only after domain restriction.

5 Polynomials

5.1 The objects and their degree

Definition 23 (Polynomial and degree) A **polynomial** is an expression

$$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0,$$

with coefficients a_i and $a_n \neq 0$. Its **degree** $\deg P = n$ is the highest power present; a_n is the **leading coefficient**. Degrees add under multiplication: $\deg(PQ) = \deg P + \deg Q$.

5.2 Roots and the factor theorem

Definition 24 (Root) A **root** (or zero) of P is a number r with $P(r) = 0$.

Theorem 25 (Factor theorem) r is a root of P if and only if $(x - r)$ divides P, i.e. $P(x) = (x - r)Q(x)$ for some polynomial Q of degree $\deg P - 1$.

Proof. Divide P by $(x - r)$ with remainder: $P(x) = (x - r)Q(x) + c$ for a constant c (the remainder has degree below 1). Setting $x = r$ gives $P(r) = c$. So $P(r) = 0$ if and only if $c = 0$, i.e. $(x - r)$ divides P. □

This is the workhorse of root-finding: **find one root, factor it out, repeat on a smaller polynomial.**

Worked example — Factoring by finding a root. $P(x) = x^3 - 6x^2 + 11x - 6$. Testing small integers, $P(1) = 0$, so $(x - 1)$ is a factor. Dividing, $P(x) = (x - 1)(x^2 - 5x + 6) = (x - 1)(x - 2)(x - 3)$. The roots are $1, 2, 3$.

5.3 Multiplicity

Definition 26 (Multiplicity of a root) A root r has **multiplicity** k if $(x - r)^k$ divides P but $(x - r)^{k+1}$ does not. A root of multiplicity 1 is **simple**; one of multiplicity 2 is **double**, and so on. Counted **with multiplicity**, a degree-n polynomial has at most n real roots — and, as we will see in Chapter 10, **exactly** n complex roots.

Worked example — A double root touches without crossing. $P(x) = (x - 2)^2(x + 1)$ has a double root at 2 and a simple root at -1. Near a double root the graph **touches** the axis and turns back (like a parabola's vertex); near a simple root it **crosses**. The multiplicity is visible in the geometry.

5.4 Vieta's formulas

The coefficients of a polynomial secretly encode symmetric combinations of its roots.

Theorem 27 (Vieta's formulas) If $P(x) = a_n x^n + \cdots + a_0$ has roots $r_1, \ldots, r_n$ (with multiplicity, possibly complex), then

$$\sum_i r_i = -\frac{a_{n-1}}{a_n}, \qquad \prod_i r_i = (-1)^n \frac{a_0}{a_n},$$

and the intermediate symmetric sums equal the intermediate coefficients with alternating signs.

For the quadratic $ax^2 + bx + c$ with roots r_1, r_2 this is the familiar

$$r_1 + r_2 = -\frac{b}{a}, \qquad r_1 r_2 = \frac{c}{a}.$$

Worked example — Using Vieta without solving. The roots of $x^2 - 7x + 10$ sum to 7 and multiply to 10. You can read off 5 and 2 by inspection — no quadratic formula needed. Vieta turns factoring into a guessing game with two clues.

Common pitfall Vieta's product formula carries the sign $(-1)^n$ and every formula is divided by the leading coefficient a_n. For a non-monic polynomial, forgetting to divide by a_n is a frequent slip: the roots of $2x^2 - 6x + 4$ sum to $\frac{6}{2} = 3$, not 6.

Exercises

18. Factor $x^3 - x^2 - x + 1$ completely and give the multiplicity of each root.
19. Without solving, find the sum and product of the roots of $3x^2 + 5x - 2$, then check by factoring.
20. A monic cubic has roots $1, 1, -2$. Reconstruct it via Vieta and expand to verify.
21. Show that if a polynomial with integer coefficients has a rational root $\frac{p}{q}$ in lowest terms, then $p \mid a_0$ and $q \mid a_n$ (the rational root theorem). Use it to find a root of $2x^3 - 3x^2 - 8x + 3$.

Chapter takeaways. Degree is the top power and adds under multiplication. By the factor theorem r is a root iff $(x - r)$ divides P, so finding one root reduces the degree. Multiplicity counts how many times a factor repeats and shows in the graph (touch vs cross). Vieta's formulas link symmetric functions of the roots to the coefficients — sum $= -\frac{a_{n-1}}{a_n}$, product $= (-1)^n \frac{a_0}{a_n}$.

6 Sequences

6.1 Counting toward a limit

A **sequence** is an infinite ordered list $a_1, a_2, a_3, \ldots$ — formally a function $\mathbb{N} \to \mathbb{R}$, written (a_n). The whole subject asks one question: **as n grows without bound, does a_n settle toward**

a single number? This is the first place the word "limit" gets a precise meaning, and it is the rehearsal for the limits of functions that follow.

6.2 Two families you must know

Definition 28 (Arithmetic and geometric sequences)

- An **arithmetic** sequence adds a fixed **common difference** d each step: $a_n = a_1 + (n-1)d$.
- A **geometric** sequence multiplies by a fixed **common ratio** r each step: $a_n = a_1 r^{n-1}$.

Worked example — Their long-run behaviour. Arithmetic $3, 5, 7, 9, \dots$ $(d = 2)$ marches to $+\infty$. Geometric $1, \frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \dots$ $(r = \frac{1}{2})$ shrinks toward 0, while $1, 2, 4, 8, \dots$ $(r = 2)$ explodes. A geometric sequence converges precisely when $|r| < 1$ (to 0), or trivially when $r = 1$.

6.3 Convergence: the ε-definition

Here is the definition that founds all of analysis. Read it with the quantifier-parsing skill from Chapter 2.

Definition 29 (Convergence of a sequence) The sequence (a_n) **converges** to the limit L, written $\lim_{n\to\infty} a_n = L$, if

$$\forall \varepsilon > 0,\ \exists N \in \mathbb{N},\ \forall n \geq N:\ |a_n - L| < \varepsilon.$$

In words: **no matter how small a tolerance ε you demand, from some index N onward every term is within ε of L.** A sequence with no such L **diverges**.

The logic is a game. An adversary names a tolerance ε (how close you must get); you must produce a threshold N (how far out you must go) past which **all** terms are inside the band $(L-\varepsilon, L+\varepsilon)$. If you can always answer, the limit holds.

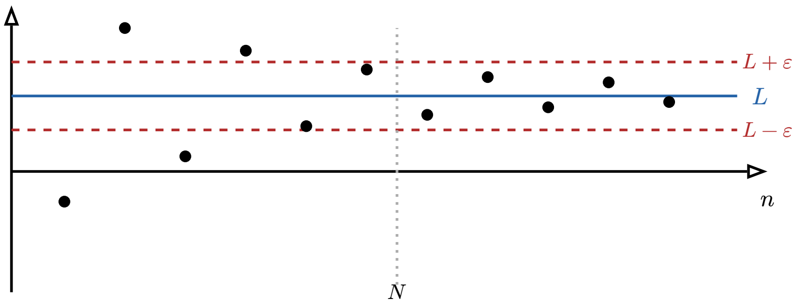

Figure 3: Convergence to L. Given the ε-band (red), there is a threshold N beyond which **every** term lands inside the band. Finitely many early terms may stray; only the eventual behaviour matters.

Worked example — An ε-proof. Claim: $a_n = \frac{1}{n} \to 0$. Given $\varepsilon > 0$, choose any integer $N > \frac{1}{\varepsilon}$. Then for $n \geq N$, $\left|\frac{1}{n} - 0\right| = \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. The adversary's ε was arbitrary, so the limit is 0.

Remark We state the ε-definition for **conceptual literacy** — so that the word "limit" rests on something solid, and so the convergence statements in later courses are not magic. You are not expected to make intricate ε-proofs the centre of your assessment; you **are** expected to read the definition and see why simple limits hold.

6.4 The monotone convergence theorem

Often we can guarantee a limit exists **without computing it**, using the completeness of $\mathbb{R}$ from Chapter 3.

Theorem 30 (Monotone convergence) A sequence that is **increasing and bounded above** converges (to its supremum). Likewise a **decreasing and bounded below** sequence converges (to its infimum).

This is completeness wearing work clothes: a sequence climbing steadily but unable to pass a ceiling has nowhere to go but toward a limit — and $\mathbb{R}$'s lack of gaps guarantees that limit is a genuine real number.

> **Worked example — A recursively defined sequence.** Let $a_1 = 1$ and $a_{n+1} = \sqrt{2 + a_n}$. One checks $a_n < 2$ for all n (bounded above) and $a_{n+1} > a_n$ (increasing). By monotone convergence the limit L exists, and passing to the limit in the recursion gives $L = \sqrt{2 + L}$, so $L^2 - L - 2 = 0$, $L = 2$. We found the limit **because** we first knew it existed.

Common pitfall Boundedness alone does not give convergence: $a_n = (-1)^n$ is bounded but oscillates forever between -1 and 1. **Monotonicity is essential** — it forbids oscillation. And convergence alone does not require monotonicity; the theorem is a sufficient condition, not a necessary one.

Exercises

22. Give the explicit formula for the arithmetic sequence with $a_1 = 5$, $d = -3$, and decide whether it converges.
23. For which r does $a_n = r^n$ converge, and to what?
24. Prove from the definition that $a_n = \frac{2n+1}{n} \to 2$.
25. Let $a_1 = 2$, $a_{n+1} = \frac{a_n}{2} + \frac{1}{a_n}$. Show it is decreasing and bounded below by $\sqrt{2}$, hence converges, and find the limit.

Chapter takeaways. A sequence is a list indexed by $\mathbb{N}$; arithmetic adds d, geometric multiplies by r (converging iff $|r| < 1$). Convergence to L means: for every ε there is an N past which all terms are within ε of L. The monotone convergence theorem (a face of completeness) guarantees a limit for bounded monotone sequences even when you cannot yet compute it.

7 Limits and Continuity

7.1 From sequences to functions

We now ask of a **function** what we asked of a sequence: as the input x approaches a point a, does the output $f(x)$ approach a single value? The cleanest way to say so reuses sequences directly.

Definition 31 (Limit of a function (sequential criterion)) $\lim_{x\to a} f(x) = L$ means: for **every** sequence (x_n) in the domain with $x_n \to a$ (and $x_n \neq a$), the image sequence satisfies $f(x_n) \to L$. Equivalently, in ε–δ form,

$$\forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x:\ 0 < |x-a| < \delta \Rightarrow |f(x) - L| < \varepsilon.$$

The sequential criterion is a powerful **disproof** tool: to show a limit does **not** exist, exhibit two input sequences approaching a whose outputs head to different places.

> **Worked example — A limit that fails to exist.** Let $f(x) = \sin\left(\frac{1}{x}\right)$ near 0. Along $x_n = \frac{1}{n\pi}$ we get $f(x_n) = 0$; along $x_n = \frac{1}{2\pi n + \frac{\pi}{2}}$ we get $f(x_n) = 1$. Two input sequences both tending to 0, two different output limits — so $\lim_{x\to 0} f(x)$ does not exist.

7.2 Continuity

Definition 32 (Continuity) f is **continuous at a** if a is in the domain and

$$\lim_{x\to a} f(x) = f(a).$$

It is **continuous on a set** if continuous at each of its points. Intuitively: the value the function **approaches** equals the value it **takes** — no jump, no hole.

The ε–δ unfolding is the very sentence we parsed in Chapter 2: **for every output tolerance ε, there is an input tolerance δ so that inputs within δ of a have outputs within ε of $f(a)$.** Keep inputs close enough and outputs stay as close as you like.

Proposition 33 (Continuity of standard functions) Polynomials, exp, ln (on $(0, \infty)$), sin, cos, and $|\cdot|$ are continuous everywhere on their domains. Sums, products, quotients (where the denominator is non-zero), and compositions of continuous functions are continuous. So essentially every function written by a single formula is continuous wherever it is defined.

Common pitfall Continuity can fail in three ways at a: a **jump** (left and right limits differ), a **hole** (limit exists but $f(a)$ is missing or wrong), or a **blow-up** (the limit is infinite). The function $f(x) = \frac{1}{x}$ is continuous on its whole domain $\mathbb{R} \setminus \{0\}$ — it is not "discontinuous at 0" because 0 is simply not in the domain. Continuity is only ever discussed at points of the domain.

7.3 Two theorems that need completeness

Continuous functions on closed bounded intervals have two powerful guarantees. Both rest quietly on the completeness of $\mathbb{R}$.

Theorem 34 (Intermediate Value Theorem (IVT)) If f is continuous on $[a, b]$ and y lies between $f(a)$ and $f(b)$, then $f(c) = y$ for some $c \in [a, b]$. In particular, if $f(a)$ and $f(b)$ have **opposite signs**, f has a root in (a, b).

A continuous graph drawn from height $f(a)$ to height $f(b)$ cannot teleport past an intermediate height — it must cross it. The "no gaps" of $\mathbb{R}$ is what forbids the teleport.

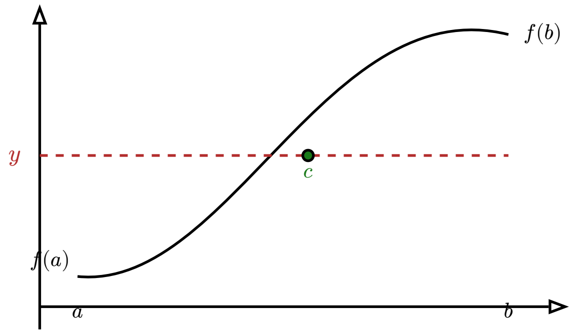

Figure 4: The IVT. A continuous curve rising from $f(a)$ to $f(b)$ must pass through every intermediate height y — there is a c with $f(c) = y$. With opposite signs at the ends, one such height is 0: a guaranteed root.

Worked example — Locating a root with the IVT. $f(x) = x^3 + x - 1$ is continuous, with $f(0) = -1 < 0$ and $f(1) = 1 > 0$. By the IVT it has a root in $(0, 1)$ — even though no nice formula produces it. Repeatedly bisecting the interval (sign-checking the midpoint) corners the root to any precision: this is the **bisection method**.

Theorem 35 (Extreme Value Theorem (EVT)) If f is continuous on a **closed bounded** interval $[a, b]$, then f attains a maximum and a minimum value on $[a, b]$ — there exist points where f is largest and smallest.

Common pitfall Both hypotheses are essential. On the **open** interval $(0, 1)$, $f(x) = x$ attains neither a max nor a min. On the **unbounded** $[0, \infty)$, $f(x) = x$ has no max. Drop "closed" or "bounded" and the guarantee evaporates — a fact we will lean on when hunting extrema in the next chapter.

Exercises

26. Show $\lim_{x\to 0} f(x)$ does not exist for $f(x) = \frac{|x|}{x}$ by using two sequences.
27. Use the IVT to prove $\cos x = x$ has a solution in $(0, 1)$.
28. Where is $f(x) = \frac{x^2-1}{x-1}$ continuous, and what value at $x = 1$ would remove the discontinuity?
29. Give a continuous function on $(0, 1]$ with no maximum, and explain which EVT hypothesis fails.

Chapter takeaways. A function limit can be tested through sequences: $f(x_n) \to L$ for **every** $x_n \to a$; two sequences with different output limits kill the limit. Continuity at a means $\lim_{x\to a} f(x) = f(a)$, and standard formula-functions are continuous on their domains. On closed bounded intervals, continuity buys the IVT (intermediate values, hence roots) and the EVT (attained max and min) — both resting on completeness.

8 Derivatives

8.1 The slope of a curve, made precise

How fast is f changing at $x = a$? Average the change over a small step h and shrink the step. The average rate $\frac{f(a+h)-f(a)}{h}$ is the slope of the **secant** line through two nearby points; as $h \to 0$ the secant pivots into the **tangent**, and its slope is the derivative.

Definition 36 (Derivative) f is **differentiable at a** if the limit

$$f'(a) = \lim_{h\to 0} \frac{f(a+h) - f(a)}{h}$$

exists; that number is the **derivative**, the slope of the tangent line at $(a, f(a))$, and the instantaneous rate of change of f there.

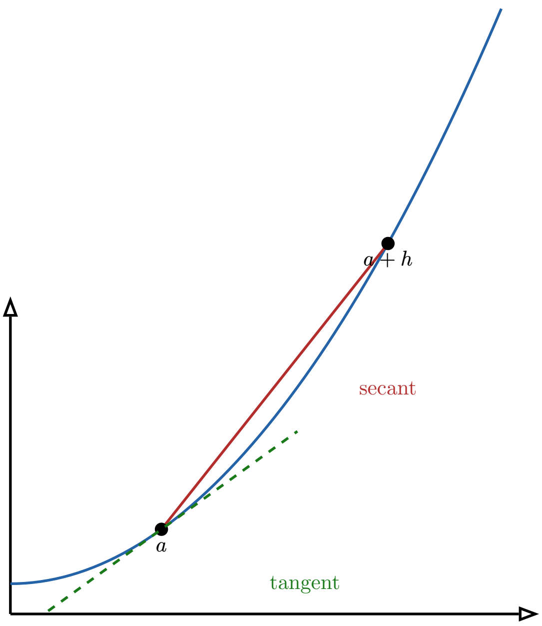

Figure 5: As $h \to 0$ the secant (red) through $(a, f(a))$ and $(a+h, f(a+h))$ rotates toward the tangent (green). The derivative $f'(a)$ is the tangent's slope.

Theorem 37 (Differentiable implies continuous) If f is differentiable at a, it is continuous at a. The converse fails: $f(x) = |x|$ is continuous at 0 but has no derivative there — the graph has a **corner**, and the secant slopes approach -1 from the left and $+1$ from the right.

8.2 The differentiation rules

Proposition 38 (Rules of differentiation) For differentiable f, g and constant c:

$$\begin{aligned} (f+g)' &= f' + g', \quad (cf)' = cf', \\ (fg)' &= f'g + fg' && \text{(product rule)}, \\ \left(\frac{f}{g}\right)' &= \frac{f'g - fg'}{g^2} && \text{(quotient rule, } \; g \neq 0 \text{).} \end{aligned}$$

The basic library: $(x^n)' = nx^{n-1}$, $(e^x)' = e^x$, $(\ln x)' = \frac{1}{x}$, $(\sin x)' = \cos x$, $(\cos x)' = -\sin x$.

Common pitfall The product rule is **not** $f'g'$ and the quotient rule's numerator order matters — it is $f'g - fg'$, not $fg' - f'g$. A sign error here propagates through an entire problem. When unsure, recompute from $(fg)' = f'g + fg'$, which is symmetric and easy to recall.

8.3 The chain rule

The most important rule, because real functions are built by composition.

Theorem 39 (Chain rule) If g is differentiable at x and f at $g(x)$, then $f \circ g$ is differentiable at x and

$$(f \circ g)'(x) = f'(g(x)) \cdot g'(x).$$

"Differentiate the outside at the inside, times the derivative of the inside."

Worked example — Peeling a composition. To differentiate $h(x) = \sin(x^2 + 1)$: outer $f = \sin$, inner $g = x^2 + 1$. Then $h'(x) = \cos(x^2 + 1) \cdot 2x$. For $h(x) = e^{3x}$, $h'(x) = e^{3x} \cdot 3 = 3e^{3x}$.

Common pitfall The commonest chain-rule error is forgetting the inner factor $g'(x)$. The derivative of $\sin(2x)$ is $2\cos(2x)$, **not** $\cos(2x)$. Whenever the input to a function is itself more than a bare x, the chain rule is in play.

8.4 Derivative of an inverse function

Theorem 40 (Inverse function derivative) If f is differentiable and invertible with $f'(f^{-1}(y)) \neq 0$, then

$$(f^{-1})'(y) = \frac{1}{f'(f^{-1}(y))}.$$

Proof. Differentiate the identity $f(f^{-1}(y)) = y$ with the chain rule: $f'(f^{-1}(y)) \cdot (f^{-1})'(y) = 1$. Solve for $(f^{-1})'(y)$. □

Worked example — Recovering a known derivative. Take $f = \exp$, so $f^{-1} = \ln$ and $f' = \exp$. The formula gives $(\ln)'(y) = \frac{1}{\exp(\ln y)} = \frac{1}{y}$ — the logarithm's derivative, derived rather than memorised. Similarly $\arctan'(y) = \frac{1}{1+y^2}$.

8.5 Finding and classifying extrema

Derivatives locate the high and low points the EVT promised exist.

Theorem 41 (Fermat's stationary condition) If f has a **local** maximum or minimum at an interior point c and is differentiable there, then $f'(c) = 0$. Such a c is a **critical** (or stationary) point.

Common pitfall $f'(c) = 0$ is **necessary, not sufficient**, for an extremum: $f(x) = x^3$ has $f'(0) = 0$ yet 0 is neither a max nor a min (it is an inflection). And extrema can also occur where f' **does not exist** (corners) or at the **endpoints** of the domain. To find the global max/min on $[a, b]$, check all three: critical points, non-differentiable points, and endpoints.

Theorem 42 (First and second derivative tests) Let c be a critical point of a differentiable f.

- **First derivative test:** if f' changes $+ \to -$ at c, it is a local max; if $- \to +$, a local min; if no sign change, neither.
- **Second derivative test:** if $f'(c) = 0$ and $f''(c) < 0$, local max; if $f''(c) > 0$, local min; if $f''(c) = 0$, the test is inconclusive.

Remark **Local versus global.** A local extremum is best only in its immediate neighbourhood; a global (absolute) extremum is best over the whole domain. On a closed bounded interval the global max/min are found by comparing the function's values at all the candidate points above — the EVT guarantees they are among them.

8.6 Convexity, concavity, and inflection

The **second** derivative measures how the slope itself changes — the curve's bending.

Definition 43 (Convex and concave) f is **convex** on an interval if its graph lies **above** every one of its tangent lines (it "holds water", curving upward); equivalently, if $f'' \geq 0$ there. It is **concave** if it lies **below** its tangents ($f'' \leq 0$, "spills water"). A point where convexity changes — where f'' changes sign — is an **inflection point**.

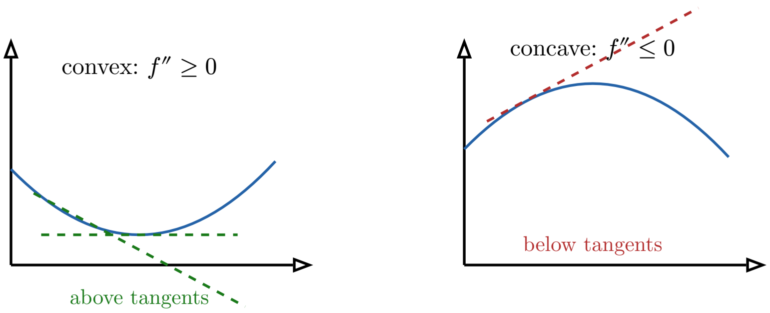

Figure 6: Convex curves lie above their tangent lines and bend upward; concave curves lie below and bend downward. The geometric reading "above tangent lines" is the working definition of convexity.

Worked example — Reading convexity from $f"$. $f(x) = x^3$ has $f''(x) = 6x$: concave on $(-\infty, 0)$, convex on $(0, \infty)$, with an inflection at 0. $f(x) = e^x$ has $f'' = e^x > 0$ everywhere — convex throughout, which is why it always curves upward.

8.7 Taylor approximations

Near a point, a differentiable function looks like a line; with the second derivative, like a parabola. Taylor approximation makes "looks like" precise and is the practical payoff of the whole chapter.

Definition 44 (Taylor polynomial) The **Taylor polynomial of order n** of f at a is

$$T_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2}(x-a)^2 + \cdots$$

It is the degree-n polynomial matching f and its first n derivatives at a. The **order-1** polynomial is the tangent line; the **order-2** is the best-fit parabola.

We **state** the general order-n form, but in practice orders 1 and 2 carry almost all the weight. The order-1 approximation $f(x) \approx f(a) + f'(a)(x-a)$ — "linearization" — is the everyday tool.

Worked example — The standard small-x approximations. At $a = 0$, three approximations recur constantly (especially in probability, where small perturbations abound):

$$e^x \approx 1 + x, \qquad \ln(1-x) \approx -x, \qquad \sin x \approx x \quad (x \text{ small}).$$

Each is just the order-1 Taylor polynomial: e.g. $\sin'(0) = \cos 0 = 1$, so $\sin x \approx \sin 0 + 1 \cdot x = x$. Adding the order-2 term sharpens them, e.g. $e^x \approx 1 + x + \frac{x^2}{2}$.

Common pitfall These approximations are only good for **small** x (near the expansion point). $e^x \approx 1 + x$ is excellent at $x = 0.01$ and useless at $x = 5$. The error grows with the distance from a and with the size of the next derivative; a remainder bound (stated in later courses) quantifies exactly how far you may trust it.

Exercises

30. Differentiate $f(x) = x^2 e^{-x}$, find its critical points, and classify each with the second derivative test.
31. Use the chain rule for $\frac{d}{dx}\ln(\cos x)$ and state where it is valid.
32. Find the intervals of convexity/concavity and the inflection point(s) of $f(x) = x^4 - 6x^2$.
33. Apply the inverse-derivative formula to $f(x) = x^3$ to compute $(f^{-1})'(8)$.
34. Approximate $\ln(1.02)$ and $e^{0.03}$ with order-1 Taylor, then with order-2, and compare.

Chapter takeaways. The derivative is the limit of secant slopes — the tangent slope and instantaneous rate. Differentiability implies continuity, not conversely (corners). Master the product, quotient, chain, and inverse-function rules. Critical points ($f' = 0$) are candidate extrema, classified by the first/second derivative tests and compared with endpoints and corners for global extrema. The sign of f'' gives convexity (above tangents) vs concavity, with inflections where it switches. Taylor polynomials approximate f near a; orders 1 and 2 — and $e^x \approx 1 + x$, $\ln(1-x) \approx -x$, $\sin x \approx x$ — are the daily tools.

9 Integration

9.1 Accumulation, and a surprising connection

Differentiation measures **rate**; integration measures **accumulation** — total distance from a speed, total area under a curve. The definite integral $\int_a^b f(x)\,dx$ is the (signed) area between the graph of f and the axis from a to b, built as a limit of thin rectangles. The astonishing fact, the climax of elementary calculus, is that this accumulation is undone by differentiation.

Definition 45 (Antiderivative) An **antiderivative** of f is a function F with $F' = f$. Any two antiderivatives differ by a constant, so we write the **indefinite integral**

$$\int f(x)\,\mathrm{d}x = F(x) + C.$$

9.2 The Fundamental Theorem of Calculus

Theorem 46 (Fundamental Theorem of Calculus) Let f be continuous on $[a, b]$.

1. The **accumulation function** $G(x) = \int_a^x f(t)\,\mathrm{d}t$ is an antiderivative of f: $G'(x) = f(x)$.
2. (**Evaluation**) If F is **any** antiderivative of f, then

$$\int_a^b f(x)\,\mathrm{d}x = F(b) - F(a).$$

Part 1 says differentiation and integration are inverse processes; Part 2 turns the hard limit-of-rectangles into easy arithmetic: find an antiderivative, subtract its values at the endpoints.

Worked example — Area under a parabola. $\int_0^1 x^2\,\mathrm{d}x$. An antiderivative of x^2 is $\frac{x^3}{3}$, so the area is $\left[\frac{x^3}{3}\right]_0^1 = \frac{1}{3} - 0 = \frac{1}{3}$. No rectangles required — the FTC did the work.

Common pitfall Don't forget the constant $+C$ in **indefinite** integrals, and don't carry it into **definite** ones — it cancels in $F(b) - F(a)$. Also, the FTC's evaluation form requires f continuous on $[a, b]$; applying it across a point where f blows up (e.g. $\int_{-1}^1 \frac{1}{x^2}\,\mathrm{d}x$) gives nonsense.

9.3 Integration by parts

Reverse the product rule and you get the most important integration technique.

Theorem 47 (Integration by parts)

$$\int u(x)v'(x)\,\mathrm{d}x = u(x)v(x) - \int u'(x)v(x)\,\mathrm{d}x.$$

In differential shorthand, $\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u$.

Proof. Integrate the product rule $(uv)' = u'v + uv'$ over the interval and rearrange. □

The art is choosing which factor is u (to be differentiated) and which is v' (to be integrated): pick u so that u' is **simpler**.

Worked example — Integrating $x e^x$. Let $u = x$ (so $u' = 1$, simpler) and $v' = e^x$ (so $v = e^x$). Then

$$\int xe^x\,\mathrm{d}x = xe^x - \int 1 \cdot e^x\,\mathrm{d}x = xe^x - e^x + C = (x-1)e^x + C.$$

A clever standard trick: $\int \ln x\,\mathrm{d}x$ with $u = \ln x$, $v' = 1$ gives $x \ln x - x + C$.

9.4 Substitution

Reverse the chain rule and you get ***u*-substitution**.

Theorem 48 (Integration by substitution) If $u = g(x)$ with g differentiable, then $\mathrm{d}u = g'(x)\,\mathrm{d}x$ and

$$\int f(g(x))g'(x)\,\mathrm{d}x = \int f(u)\,\mathrm{d}u.$$

For a definite integral, change the limits too: x from a to b becomes u from $g(a)$ to $g(b)$.

Worked example — Spotting the inner function. $\int 2x\cos(x^2)\,\mathrm{d}x$. Put $u = x^2$, $\mathrm{d}u = 2x\,\mathrm{d}x$, which is exactly present. The integral becomes $\int \cos u\,\mathrm{d}u = \sin u + C = \sin(x^2) + C$. The substitution works because the integrand already contains the derivative of the inner function.

Common pitfall When substituting in a **definite** integral, either convert the limits to the new variable (cleanest) **or** convert back to x before plugging in the original limits — but never plug the old x-limits into a u-expression. Mixing the two is a frequent and costly slip.

Exercises

35. Evaluate $\int_1^2 (3x^2 - 2x)\,\mathrm{d}x$ using the FTC.
36. Compute $\int x\sin x\,\mathrm{d}x$ by parts.
37. Compute $\int \frac{x}{x^2+1}\,\mathrm{d}x$ by substitution.
38. Find $\int_0^{\frac{\pi}{2}} \cos x\,\mathrm{d}x$, and explain geometrically why the answer is 1.
39. Use parts twice to evaluate $\int x^2 e^x\,\mathrm{d}x$.

Chapter takeaways. The definite integral accumulates signed area. An antiderivative reverses differentiation, and the Fundamental Theorem links the two: $\int_a^b f = F(b) - F(a)$ for any antiderivative F. Integration by parts reverses the product rule ($\int u\,\mathrm{d}v = uv - \int v\,\mathrm{d}u$, choose u to simplify on differentiating); substitution reverses the chain rule (set $u = g(x)$, and remember to change the limits).

10 Complex Numbers

10.1 Inventing a number to solve $x^2 = -1$

No real number squares to -1, so the equation $x^2 + 1 = 0$ has no real solution. Rather than stop, mathematics **adjoins** a new number i with $i^2 = -1$ and asks what arithmetic results. The answer is the **complex numbers** $\mathbb{C}$ — a system in which **every** polynomial equation has a solution, and in which exponentials and trigonometry turn out to be the same thing.

Definition 49 (Complex numbers, algebraic form) A **complex number** is $z = a + bi$ with $a, b \in \mathbb{R}$ and $i^2 = -1$. Here $a = \mathfrak{R}(z)$ is the **real part** and $b = \mathfrak{I}(z)$ the **imaginary part**. Addition is componentwise; multiplication uses $i^2 = -1$:

$$(a + bi)(c + di) = (ac - bd) + (ad + bc)i.$$

The **conjugate** is $\overline{z} = a - bi$, and $z\overline{z} = a^2 + b^2$ is real and non-negative.

The conjugate is the key to division: to compute $\frac{z}{w}$, multiply top and bottom by $\overline{w}$ to make the denominator real.

Worked example — Dividing complex numbers.

$$\frac{2+i}{1-i} = \frac{(2+i)(1+i)}{(1-i)(1+i)} = \frac{2+2i+i+i^2}{1+1} = \frac{1+3i}{2} = \frac{1}{2} + \frac{3}{2}i.$$

10.2 The complex plane and polar form

Plotting $z = a + bi$ as the point (a, b) turns $\mathbb{C}$ into a plane. A point is equally described by its distance from the origin and its angle — **polar form**.

Definition 50 (Modulus, argument, polar form) The **modulus** $|z| = \sqrt{a^2 + b^2}$ is the distance from the origin; the **argument** $\arg(z) = \theta$ is the angle from the positive real axis. Then

$$z = r(\cos\theta + i\sin\theta), \qquad r = |z|.$$

The argument is defined only up to adding multiples of 2π.

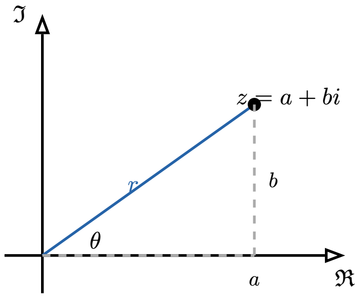

Figure 7: A complex number as a point in the plane. Its modulus r is the distance to the origin, its argument θ the angle to the real axis; $a = r\cos\theta$, $b = r\sin\theta$.

Polar form makes multiplication geometric: **moduli multiply and arguments add.** If $z_1 = r_1(\cos\theta_1 + i\sin\theta_1)$ and likewise z_2, then $z_1 z_2 = r_1 r_2(\cos(\theta_1 + \theta_2) + i\sin(\theta_1 + \theta_2))$. Multiplying by a complex number **rotates and scales** the plane.

10.3 Euler's formula

The addition law for arguments is exactly the law of exponents — a coincidence too perfect to be one.

Theorem 51 (Euler's formula) For real θ,

$$e^{i\theta} = \cos\theta + i\sin\theta.$$

Hence every complex number has the compact polar form $z = re^{i\theta}$, and multiplication reads $r_1 e^{i\theta_1} \cdot r_2 e^{i\theta_2} = r_1 r_2 e^{i(\theta_1 + \theta_2)}$.

Setting $\theta = \pi$ yields **Euler's identity** $e^{i\pi} + 1 = 0$, binding e, i, π, 1, and 0 in a single line — often called the most beautiful equation in mathematics. More practically, $e^{i\theta}$ recovers the addition formulas of Chapter 4 for free: expand $e^{i(\alpha+\beta)} = e^{i\alpha}e^{i\beta}$ and match real and imaginary parts.

> **Worked example — Powers via De Moivre.** Since $z = re^{i\theta}$, $z^n = r^n e^{in\theta} = r^n(\cos n\theta + i \sin n\theta)$ — **De Moivre's formula**. To compute $(1+i)^8$: here $r = \sqrt{2}$, $\theta = \frac{\pi}{4}$, so $(1+i)^8 = \left(\sqrt{2}\right)^8 e^{i \cdot 8\frac{\pi}{4}} = 16e^{i2\pi} = 16$. Polar form turns an eighth power into one multiplication.

10.4 Solving polynomial equations over $\mathbb{C}$

Theorem 52 (Fundamental Theorem of Algebra) Every non-constant polynomial with complex coefficients has at least one complex root; consequently a degree-n polynomial factors completely as $a_n(x - r_1)\cdots(x - r_n)$ over $\mathbb{C}$ and has **exactly** n roots counted with multiplicity.

This is why $\mathbb{C}$ is the natural home of polynomials: the holes that $\mathbb{R}$ left ($x^2 + 1 = 0$) are all filled. Roots are found cleanly in polar form.

> **Worked example — The cube roots of unity.** Solve $z^3 = 1$. Write $1 = e^{i2\pi k}$ for integer k, so $z = e^{i2\pi\frac{k}{3}}$ for $k = 0, 1, 2$:
>
> $$z = 1, \quad e^{i2\frac{\pi}{3}} = -\frac{1}{2} + \frac{\sqrt{3}}{2}i, \quad e^{i4\frac{\pi}{3}} = -\frac{1}{2} - \frac{\sqrt{3}}{2}i.$$
>
> The three roots sit at the vertices of an equilateral triangle on the unit circle — a pattern that generalises: the n solutions of $z^n = 1$ are equally spaced around the circle.

Common pitfall When taking n-th roots, you must let the argument range over all of $k = 0, 1, \ldots, n-1$ to capture **all** n roots. Writing only $\arg = 0$ finds a single root and misses the rest — the most common error in solving $z^n = w$.

Exercises

40. Compute $(3 - 2i) + (1 + 4i)$, $(3 - 2i)(1 + 4i)$, and $\frac{3-2i}{1+4i}$.
41. Write $z = -1 + i$ in polar form and compute z^6 via De Moivre.
42. Use Euler's formula to derive $\cos(2\theta) = \cos^2\theta - \sin^2\theta$.
43. Find all complex solutions of $z^2 = -4$ and of $z^4 = 1$, and plot them.
44. Factor $x^2 + x + 1$ completely over $\mathbb{C}$ (note its roots are two of the cube roots of unity).

Chapter takeaways. Complex numbers $a + bi$ ($i^2 = -1$) form a field; conjugation enables division. Plotted in the plane, each has a modulus and argument, and polar form $z = re^{i\theta}$ via Euler's formula ($e^{i\theta} = \cos\theta + i\sin\theta$) makes multiplication "multiply moduli, add arguments" — rotation and scaling. De Moivre powers and roots follow, and the Fundamental Theorem of Algebra guarantees a degree-n polynomial has exactly n roots over $\mathbb{C}$, equally spaced for $z^n = w$.

11 Closing

You began this course already able to **compute**; you leave it able to **justify**. Every chapter took an object you half-knew and gave it a precise definition, a theorem stating exactly what is true of it, and a proof saying why. The logic of Chapter 1 let you read those theorems; the real numbers of Chapter 3 gave them a gapless stage; functions, polynomials, and sequences supplied the actors; and the calculus — limits, derivatives, integrals — and the complex numbers completed the classical toolkit.

What carries forward is not a list of formulas but three habits: **state precisely** what you are claiming, **check the hypotheses** before invoking a theorem, and **bound what you cannot compute exactly**. Every later course in mathematics, statistics, and machine learning is written in this language. You can now read it — and, increasingly, write it.

Course takeaways. Mathematics is precise statements, justified. Logic and set notation are the language; the complete ordered field $\mathbb{R}$ is the stage; functions, polynomials, and sequences are the first objects; limits, continuity, derivatives, convexity, Taylor approximation, and integration are the calculus; and $\mathbb{C}$ with Euler's formula completes the number system. Hypotheses matter, contrapositives are free, and what cannot be computed can often be bounded.